

- 8 A star in a distant galaxy consists mostly of hydrogen. Visible light emitted by the star consists of discrete spectral absorption lines, as represented in Figure 8.1.

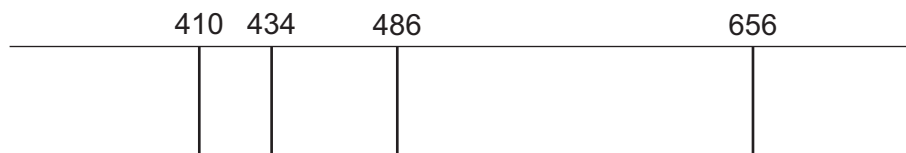


Figure 8.1

The numbers represent the wavelengths, in nm, of the spectral lines.

- (a) Explain, with reference to photons, how the existence of these spectral lines provides evidence for discrete electron energy levels in the hydrogen atom.

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..... [3]

- (b) The galaxy is at a distance of 6.7×10^{24} m from the Earth.

- (i) Explain why the spectral lines observed on the Earth are at different wavelengths from those shown in Figure 8.1.

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..... [2]

- (ii) On Figure 8.1, draw possible positions for the four spectral lines observed on the Earth. [2]

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(iii) The Hubble constant may be taken as $2.1 \times 10^{-18} \text{ s}^{-1}$.

Determine the observed wavelength, in nm, of the lowest frequency spectral line of visible light from the star.

wavelength = nm [3]

[Total: 10]